

Pragmatic Theories Enhance Understanding of Implied Meanings in LLMs

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Abstract

The ability to accurately interpret implied meanings plays a crucial role in human communication and language use, and language models are also expected to possess this capability. This study demonstrates that providing language models with pragmatic theories as prompts is an effective approach for tasks to understand implied meanings. Specifically, we propose an approach in which an overview of pragmatic theories, such as Gricean pragmatics and Relevance Theory, is presented as a prompt to the language model, guiding it through a step-by-step reasoning process to derive a final interpretation. Experimental results showed that, compared to the baseline, which prompts intermediate reasoning without presenting pragmatic theories (0-shot Chain-of-Thought), our methods enabled language models to achieve up to 9.6% higher scores on pragmatic reasoning tasks. Furthermore, we show that even without explaining the details of pragmatic theories, merely mentioning their names in the prompt leads to a certain performance improvement (around 1-3%) in larger models.

1 Introduction

Language often contains implicit meanings, unstated intentions, and context-dependent interpretations, collectively referred to as **implied meanings**. The ability to correctly understand and interpret implied meanings plays a crucial role in human communication (Grice, 1989; Levinson, 1983; Carston, 2002). In order to correctly interpret the implied meanings, people use background knowledge of communication, such as dialogue context, common sense, and cultural background. Using these, they achieve effective communication in their daily lives. The ability to understand implied meanings

is important not only for humans but also for AI systems or robots based on them. For example, when a human’s intention is ambiguous, inferring their intentions and taking actions proactively while considering the situation and common sense is necessary (Tanaka et al., 2024). The ability to interpret implied meaning, as well as understanding the nuances and intentions of language, is essential for AI and robots to be needed in human society.

In recent years, following the success of Large Language Models (LLMs), research has been conducted on employing them to develop systems that have abilities to interpret language and situations (Ahn et al., 2022; Ma et al., 2025). Numerous benchmark tasks and datasets have been proposed to assess these capabilities (Jeretic et al., 2020; Zheng et al., 2021; Takayama et al., 2021; Hu et al., 2023; Li et al., 2024a; Sravanthi et al., 2024; Yerukola et al., 2024; Yue et al., 2024), facilitating comparisons between current language models and human performance, as well as analyses of the challenges these models face.

Two major approaches have been explored to enhance the understanding of implied meanings in LLMs: post-training involving parameter updates (Wu et al., 2024; Sravanthi et al., 2024), and in-context learning (ICL), which draws out specific capabilities of LLMs through carefully designed prompts (Yerukola et al., 2024; Ruis et al., 2024; Kim et al., 2023). ICL is particularly important, as it enables the full utilization of the model’s abilities without additional training costs, and has been addressed with extensive approaches regarding its generalization capabilities. In the context of understanding implied meanings, there is a growing need to explore not only instance-specific methods but also more instance-agnostic methods.

In this study, we propose a method that enhances model performance on tasks to understand implied meanings in a zero-shot setting without relying on prompts specific to particular problem formats

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or providing top-down hints from correct answers. Implied meanings are handled in the field of linguistics known as **pragmatics**, where various theories have been proposed regarding their properties and the mechanisms of their interpretation. The proposed method inserts summaries of theoretical frameworks from linguistic pragmatics into the model’s prompt and instructs the model to generate intermediate reasoning processes by following those theories. It is known that existing LLMs have a variety of knowledge in their parameters; however, making them recall and use this knowledge appropriately for the task is still challenging. By giving the rough outline of pragmatic theories as a bootstrap, we expected that these models could recall the related knowledge to be used and manipulate them for solving such tasks to understand the implied meanings.

Experimental results demonstrated that the proposed method consistently improved model performance on tasks to understand implied meanings without providing instance-dependent information, achieving an accuracy improvement of up to 0.096 in the experimental tasks. We tested the proposed method in both commercial-closed and open models and found that the proposed method contribute to a wide range of models. Additionally, slight score improvements were observed in many models even when the prompt did not include an overview of pragmatic theories but only mentioned their names while encouraging reasoning by following them. These investigations not only have engineering utility and can be expected to apply to upstream tasks such as dialogue, but also lead to clarifying the nature of the task of pragmatic understanding itself and what kind of thought processes are effective for executing such tasks. Our contributions can be summarized as follows:

- We proposed a **simple method that incorporates summaries of pragmatic theories (namely Gricean and Relevance Theories) into prompts** and showed that this improves LLM performance on implied meaning understanding tasks without preparing task-dependent prompts.
- We showed that even without explaining the theories in detail, **simply referencing theory names** in zero-shot Chain-of-Thought prompting leads to performance improvements over baseline methods.
- Through detailed quantitative and qualitative

analyses, we demonstrated that our method is particularly effective for tasks involving utterances that flout Grice’s maxims and for interpreting irony.

2 Related Works

2.1 Pragmatic Reasoning by Language Models

Previous research has demonstrated that methods such as post-training, Chain-of-Thought (CoT) reasoning, and few-shot learning effectively enhance the pragmatic inference capabilities of language models. As post-training approaches using pragmatic reasoning datasets, policy optimization methods such as Direct Preference Optimization (DPO) are more effective than supervised fine-tuning (Wu et al., 2024). It has also been confirmed that instruction-tuned models achieve higher scores in pragmatic reasoning tasks compared to their corresponding base models (Ruis et al., 2024; Sravanthi et al., 2024). These methods involve updating the model’s parameters.

For in-context learning methods that do not involve updating the model’s parameters, previous research has shown that CoT prompting, where guidances for correct interpretation are included in the prompt (Yerukola et al., 2024), and few-shot prompting, where the prompts provide examples of problems and correct answers similar to the target task (Ruis et al., 2024), are effective approaches. As a method that combines these approaches, experiments have reported that providing reasoning steps based on Gricean theory (Grice, 1989) within few-shot examples can serve as effective guidance for correct interpretation (Kim et al., 2023).

However, there has been insufficient research on methods for enhancing the pragmatic reasoning capabilities of language models without ad hoc interventions, such as modifying model parameters or providing top-down hints specific to the task. For large-scale pretrained LLMs, conducting post-training with parameter updates (e.g., supervised fine-tuning or preference optimization) is not only costly and labor-intensive but also carries the risk of “catastrophic forgetting,” where the performance on previously learned tasks deteriorates after fine-tuning on a specific task (Kirkpatrick et al., 2017; Li et al., 2024b). Additionally, existing in-context learning methods should be developed in light of the fact that pragmatic inference is often not an end goal in itself but rather a necessary component

ability for higher-level tasks or objectives. Given this, the approaches in prior studies (Yerukola et al., 2024; Ruis et al., 2024; Kim et al., 2023), which rely on top-down and ad hoc applications of few-shot learning or CoT prompting tailored to specific pragmatic reasoning problems, may not be sufficient. We tackle their remaining challenge to maintain generalized prompts for interpreting implied meanings using pragmatic theories.

2.2 Pragmatic Theories in Linguistics

We propose a method based on two well-established pragmatic theories from the fields of linguistics and philosophy of language: Gricean pragmatics (Grice, 1989) and Relevance Theory (Sperber and Wilson, 1995).

Gricean Theory Grice proposed a pragmatic theory in which he argued that the correct interpretation of what a speaker means in an utterance is achieved by assuming that participants in a conversation adhere to, or at least appear to adhere to, the *Cooperative Principle*. This principle serves as the foundation for the reasoning made by listeners.

Cooperative Principle (Grice, 1989)

Make your conversational contribution such as is required, at the stage at which it occurs, by the accepted purpose or direction of the talk exchange in which you are engaged (p.26).

He also argued that adherence to the cooperative principle requires the observance of the four subordinate maxims, namely *Maxim of Quantity*, *Quality*, *Relation*, and *Manner* (see Appendix §A).

Grice explains the understanding and interpretation of implied meanings based on the idea that our conversations generally adhere to these maxims. When they do not, the listener infers meaning by recognizing that they are violating one or more maxims while assuming that the speaker still follows the *cooperative principle*. Grice’s theory has had a significant impact not only in linguistics and philosophy of language but also in Natural Language Processing (NLP), where it has been widely applied (Krause and Vossen, 2024).

Though Grice’s work founded modern pragmatics, later research exposed its limits and inspired refinements such as Relevance Theory, which offers a more cognitively plausible account of utterance interpretation (Levinson, 2000; Horn, 1984; Wearing, 2015).

Relevance Theory A key feature of Relevance Theory, proposed by Sperber and Wilson, is its cognitive approach to pragmatic meaning (Sperber and Wilson, 1995). Relevance Theory defines *relevance* in terms of two factors: the *cognitive effect*, which refers to the degree to which an utterance influences the listener’s thoughts, and the *processing effort* required to understand or interpret the utterance. All else being equal, an utterance is considered more relevant if it produces greater *cognitive effects* and requires less *processing effort*. Relevance Theory asserts that, based on this notion of *relevance*, we expect the *presumption of relevance* (see Appendix §B) in communication and that pragmatic meaning is interpreted accordingly.

Our study proposes a method that incorporates summaries of Gricean pragmatics and Relevance Theory into the model’s prompt, guiding the model to generate reasoning processes aligned with these theories.

2.3 Zero-shot prompt templates

It is known that specifying certain thinking methods as prompts can generically improve model performance without giving language models problem-answer pairs (Brown et al., 2020) or providing explicit hints (Wei et al., 2022). Kojima et al. (2022) showed that simply inputting the text “Let’s think step by step.” to a model can significantly improve performance on benchmarks for various tasks, and this method is called *zero-shot Chain-of-Thought*. Such methods are not only convenient for practical use of language models, but also noteworthy for exploring the foundations and mechanisms that realize their capabilities, so various studies have been conducted subsequently (Wang et al., 2023a,b; Schulhoff et al., 2025).

In such contexts, several prompt templates have been discovered that aim to maximize LLM capabilities in specific tasks or domains. For example, He et al. (2024) shows that prompt templates that make models extract keywords and topics from source sentences before performing final translation improve LLM performance on translation tasks. Additionally, Sivarakumar et al. (2024) demonstrated that zero-shot prompt templates based on medical domain knowledge are effective in clinical information extraction tasks. Methods of *jailbreak attacks* (Yi et al., 2024), which are intended to elicit harmful or inappropriate outputs from models by devising prompts, might also be positioned within this context.